

Program : Diploma in Wood and Paper Technology	
Course Code : 3126	Course Title: Wood Chemistry Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To enable the students to apply the principles of wood analysis learned in wood chemistry
- To impart basic knowledge on the qualitative analysis of inorganic salt.

Course Prerequisites:

Topic	Course code	Course name	Semester
Wood analysis		Wood Chemistry	3
Salt analysis		Applied Chemistry,	1

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Conduct the qualitative analysis of inorganic salt	15	Applying
CO2	Demonstrate the Identification of wood	3	Understanding
CO3	Conduct the Analysis of wood	17	Applying
CO4	Analyse wood preservatives	6	Applying
	Lab Exam	4	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Conduct the qualitative analysis of inorganic salt		
M1.01	Systematic analysis for the identification of cation and anion in inorganic salt Cation - ammonium, lead, aluminium, barium, calcium, magnesium, Anion - carbonate, acetate, chloride, nitrate, sulphate	15	Applying
CO2	Demonstrate the Identification of wood		
M2.01	Teak rose wood, acasia, jack wood, rubber wood, vankana, mango, aini and irul	3	Understanding
	Lab Exam – I	2	
CO3	Conduct the Analysis of wood		
M3.01	Determination of lignin in raw material	3	Applying
M3.02	Demonstrate the determination of pentosan in raw material	3	Applying
M3.03	Demonstrate the determination of alpha, beta, gamma and holocellulose	3	Applying
M3.04	Determination of moisture content of wood by oven dry method	3	Applying
M3.05	Determination of ash content of wood	3	Applying
M3.06	Determine the humidity and temperature measured by hygrometer	2	Applying

CO4	Analyse Wood Preservatives		
M4.01	To determine the concentration of a preservative solution by using hydrometer	1	Applying
M4.02	Conduct the dry salt retention (DSR) on a preservative treated specimen	2	Applying
M4.03	To determine the qualitative analysis for the depth of penetration of boron treated wood	3	Applying
	Lab Exam – II	2	

Text / Reference:

T/R	Book Title/Author
T1	Browning B. L. “The Chemistry of Wood”, John Wiley & Sons. 1981
R1	Kocurek M. J., “Pulp and Paper Manufacture, Volume 3: Secondary Fibers and Non-wood Pulping (Ed. Hamilton F. and Leopold B.)”, TAPPI Press. 1987
R2	Eckenfelder W.W. and Ford D., “Water Pollution Control” 3rd Ed., Jonkins Publishing Company, 2000
R3	Sjostrom E., “Wood Chemistry Fundamentals and Applications”, 2nd Ed., TAPPI Press.

Online Resources:

Sl.No	Website Link
1	https://www.youtube.com/watch?v=KnhnTG8p61o
2	https://www.youtube.com/watch?v=oj1Nr3MThzU
3	https://www.youtube.com/watch?v=P6TzlaHUsgs